

Introduction to Servlets

👉 What are servlets?

- **Definition:** Servlets are server-side Java programs that handle client requests and generate dynamic web content. They run within a servlet container managed by a web server.
- **Execution:** Unlike standard Java programs, servlets are not executed by the Java Virtual Machine (JVM) directly and thus do not require a main method.

👉 How Servlets Work:

1. Request Flow:

- The client (web browser) sends a request to the web server.
- The web server forwards the request to the servlet container.
- The servlet container locates the appropriate servlet to handle the request.
- The servlet processes the request, potentially interacting with a database.
- The servlet generates a response and sends it back to the web server.
- The web server sends the response to the client's web browser.

2. Servlet Container:

- A servlet container, such as Apache Tomcat, is a part of a web server that interacts with the servlets.
- It is responsible for managing the lifecycle of servlets, mapping requests to servlets, and ensuring proper request handling.

👉 Key Points:

- **No Main Method:** Since servlets are not standalone applications, they do not need a main method. Instead, they are managed by the servlet container.
- **Server Software:** To run servlets, you need server software like Apache Tomcat that can host the servlet container and handle HTTP requests.

👉 Setting Up a Server:

- **Server Software:** Install server software (e.g., Apache Tomcat).
- **Run Servlet Container:** Ensure the servlet container is running to process incoming HTTP requests and handle servlet execution.

